PATT43: The 43rd International Pupils' Attitudes Towards Technology Research Conference
Hosted by the Swedish National Centre for School Technology Education (Cetis) and Technology and Science Education Research (TekNaD), Linköping University, Norrköping, Sweden, June 15th to June 18th, 2026

Investigating the Progression of Students' Designerly Capability in Technology Education through a Grounded Theory Project

Manus McDyer[1,2], Nicolaas Blom[2] and Jeffrey Buckley[1,3]
[1]Technological University of the Shannon: Midlands Midwest
[2]University of Limerick, Ireland
[3]Linköping University, Sweden
A00302279@student.tus.ie

Abstract

Technology education strives to foster active citizens who have an appreciation and understanding of the made world and the impact of the use, modification and manipulation of technology towards positive change. Being designerly has been identified as a central attribute to this shared goal, where possessing designerly capabilities will enable active participation and engagement in society. Design holds a significant position within technology education internationally and this recognition is evident in curricula and research activity, although there still exists a lack of consensus towards a shared understanding of what it means to be a designer that designs, how designing unfolds, and how designerly capabilities can be fostered.

This paper presents a point during the second phase of a grounded theory project where the central research objective is to investigate the complexities of learning through designing with the following research question: what does the progression of designerly capability look like for students in technology education? Two technology education academics with experience in design and one design education academic were theoretically selected. Unstructured interviews were conducted and analysed according to the principles of classical grounded theory, where both open coding was utilised for this initial analysis. In this paper, key findings identify the transferability of designers, the assessment barrier and the future of design education. These insights provide further focus for future data collection of this phase, to investigate the progression of students' designerly capability.

Key Words: Classical Grounded Theory, Technology Education, Design, Designerly Capability, Higher Education

1 INTRODUCTION

Design and the act of designing can be described as an essential element to technology education internationally, through growing research agendas (McDyer et al., 2022), curriculum developments, and the view that design is essential for the achievement of an innovative society (Williams et al., 2013). The purpose of technology education is rooted in exploring the impact of the use, modification and manipulation of technology (de Vries, 2016) to foster active citizens to understand and equip them to engage in positive change in the made world. At the heart of these purposes is allowing students to foster and elicit designerly capabilities through the engagement in design activities (Baynes, 2006; Stables, 2008). However, there exists uncertainties and ambiguity surrounding a consensus of how design and designing is defined, what design should look like in practice, and how it should be taught (Atkinson, 2017). This ambiguity and the intricacies of design(ing) and researching design (McDyer et al., 2023) coupled with the broad and open ended spectrum of research into design (McDyer et al., 2022) calls for further investigation into the complexity of design. Design has been a prominent topic of inquiry from 2000-2018 (Williams, 2016; Xu et al., 2020), which still continues to be today, and will continue to be a fundamental and common area of inquiry in technology education (Williams, 2017). Ritz & Martin (2013) identified that in order for the progression of the field of technology education, one of the necessary research agendas is how should design activities with intentions for concept learning be taught by teachers. However, adding further complexity to this need, one must be considerate of the varying descriptions of design as a noun and as a verb, the designer, and the act of designing, while also being conscious of the differentiation between design in a professional capacity and design within technology education (Spendlove, 2015). This ambiguity, uncertainty and ultimately complexity of design in technology education encapsulates the essence of design. Design prepares students for the future, which of course nobody can define, but through design and designing the future can look free and open to indefinite manipulation (Mitcham, 2001), making further investigation into design(ing) a prominent need for technology education.

This paper presents findings from an on-going classical grounded theory (CGT) project where the main research question is to investigate the complexities of design in technology education. The first phase of this project identified how design is enacted in practice within Irish technology education, and demonstrated that design is complex, intricate and that teachers more strongly value supporting students engaging in design activities compared to strictly teaching about design theory and design processes (McDyer et al., 2024). The results from the first phase led to the identification of a core category (‘designer’) for further focused investigation in this current phase presented in this paper. The iterative relationship between data collection and data analysis within a CGT methodology (Walker & Myrick, 2006) allows this paper to present an insight into initial findings during this phase. This phase focuses on the following research question, ‘what does the progression of designerly capability look like for students in technology education?’.

2 METHODS

The methodology chosen for this research is classical grounded theory, founded by Glaser and Strauss (1967). This methodology utilises an iterative approach whereby data collection and

analysis takes place simultaneously and merges the qualities of quantitative research's rigor with the depth and richness of qualitative research (Walker & Myrick, 2006).

2.1 Participant Selection

Theoretical sampling was employed to identify participants that had potential to be included in the data collection, allowing this stage of data collection to be in-line with the initial concept and theory discovery that has taken place from the first phase (Glaser & Strauss, 2017). The identified group for inclusion in sampling were academics within the field of technology education or design education, that have a research focus within the remit of design or designing. Attaining a broad international perspective was also part of the inclusion criteria, where participants were included from various countries around the world. At this current stage of the project, only three participants' data have been included in this paper as this is an on-going study.

Participants included in this paper were two academics within technology education (one male from the United Kingdom and one female from Sweden) and one male academic from design education (from the United Kingdom). They had a varying range of experiences within research and as educators, with varying undergraduate degree backgrounds.

2.2 Data collection

Online unstructured interviews were conducted on Microsoft Teams by the principal investigator and ranged from 35 minutes to 64 minutes in duration. The unstructured nature of the interviews was to limit researcher bias and to allow for open directional influence of the participants' experience on the data collected. However, this phase of the project was directed to explore the core category identified within the previous phase with more focus, which is the category of 'designer' (McDyer et al., 2024).

2.3 Researcher Positionality and Ethical Considerations

Full ethical approval was granted by the supporting university, and anonymity has been ensured through providing each participant with a pseudonym and storing data in a password protected storage system. Only the principal investigator conducted the interviews for this study and has an undergraduate degree in technology education and is currently a PhD student in this field, which supported the development of a positive rapport with the participants from a similar background. The unstructured interview design allowed for a conversational style approach between researcher and participant, which supported the gathering of insightful data. It was made clear that the participant could withdraw at any stage of the interview or remove their data up until 2 weeks post interview. The frequency of codes identified within the data is not used as an indicator of validity or importance, compared to less frequent codes. Finally, the researcher's bias and preconceptions were managed through memoing, reflexivity, and a recognition of the influence this could have on the data collected.

2.4 Data Analysis

Data analysis was aligned with the classical grounded theory methodology, where open coding was conducted for this stage of the project, as described by (Kenny & Fourie, 2015). The software NVivo was used to support coding. Biases and preconceptions of the researcher were identified and separated from the study particularly during data collection and analysis. Memo writing and reflexive journaling (Steltenpohl et al., 2024) were implemented before and after interviews, and during the data analysis stage to support this, and to ensure accurate interpretation within data analysis, and to note any key insights that provide rich data for consideration. Constant comparison has been used between the data collection and open coding processes to compare and contrast with each interview's data. After each interview, the associated transcript is openly coded to allow the researcher to become familiar with the data, the codes being generated and the emergence of the core category.

3 FINDINGS

As this data collection is still on-going, the findings section only presents the open codes that have been identified from the transcripts and are not grouped conceptually or categorised. The codes offer insight into the complexities of design, designing and designer. Direct quotes have been taken from the transcripts and the name in brackets refers to the participant's assigned pseudonym to ensure anonymity.

Despite 'designer' being identified as the core category during the analysis of phase 1, there was a lot of discussion regarding design and designing as well as the activity of the designer during interviews. Therefore, from initial analysis all codes presented below do not fall into the core category of 'designer' but are valuable findings worth describing within this paper.

3.1 Design is not formulaic… but it can be

This is a combination of 2 codes that have been identified within the data. The first code labelled 'design isn't formulaic' refers to the nature of design being free from structure, procedural steps and methods, whereby the person engaging in design will go through their own interpreted process, as best they see it to come towards a final design solution, independent and free from a formulated procedural design process.

> "But also reflecting that the things that we're showing on the slides, like the double diamond, are not real processes. They're kind of a suggested… in an idealized world, if you look a bit squinty eyed, fuzz things up, you will get some kind of process like this overall, but really the important bit is the going out and coming back in and going out and coming back in and doing that multiple times" (Hugh).

Secondly 'design can be formulaic' is a code from the data that refers to the possibility of design being formulaic, through following curated steps, a specific process to design or by allowing the assessment criterion to determine the process. Participants refer to the formulisation being

through assessment driven approaches and the challenges of implementing design in classroom practice,

> "it [designing] has become more formalized, it tends to be being led by assessment" (Frank).

3.2 Assessment can be a barrier to design

The assessment approaches, the use of visible criterion and grading rubrics have been identified as placing a limiting factor on true designing. The fluid, free moving and personal expression of students' designs and design processes can be prescribed and formulated into a procedural set of steps based on the criterion that their work will be assessed on. Often referring to the design portfolio being a polished structure of designing, although with the intent to capture each students' design processes, it can ultimately be used as a deductive design process. However, the awareness of the assessment barrier to design is supported by the difficulty and ambiguity as to what are the most beneficial assessment approaches to be used, that do not take away from true design.

> "How the design process is experienced is the assessment criteria, particularly when you think about upper secondary and post 16 where they're working towards an examination. And quite often you find that the design process is framed by the assessment" (Frank).

3.3 Where does optimal learning take place?

This code presented asks the question about where the optimal learning point along a design process is. Participants talk about the need for self-reflection along a design process to fully understand the intricacies of design, and others identify the use of design theory to build a body of awareness of the complexities of the variable processes in designing. Participants identified a pivot away from the finalised design product being the encapsulation of learning, towards a focus on the 'why' or the meaning of the process towards the outcome.

> "I discuss that a lot with my students too, how to make the technology education meaningful and not put all the focus on the making" (Grace).

3.4 Future of Design Education

Key data that formulated this code identifies the need for supporting the implementation of open and creative design tasks for students in classroom practice. The current constraints to this are through the necessity of curriculum development, time allocation and teacher professional development, among others. These implementation challenges can often discourage intentionally open and ill-defined design tasks to become closed, prescribed and formulaic to allow engagement and realisation in the classroom. The complexities of allowing the students to be autonomous and to be engaging in uncertain or ill-defined design tasks require a deeper investigation into design theory and processes, and pedagogical approaches to support this.

> "I think that's one part that I wish I could put more time on, to teach them how to be creative" (Grace).

3.5 Design is to be comfortable in uncertainty

A common trend across interviews was that of the need for students to be comfortable within uncertainty and to be confident in their own self-belief. It is discussed how engaging in design tasks bring different hurdles, challenges and unknowns which amount to uncertainty, so it is identified that students must develop their comfort in being uncomfortable in these moments.

> "To give them as much kind of confidence to tackle the other areas which are more uncertain" (Hugo).

3.6 Transferability of Designers

It is frequently discussed that design within different contexts and disciplines such as product design, engineering design and artistic design are vastly different due to the context specific skills, knowledge and experience that is required to successfully design within the associated context. What this code refers to however, are the elements of the designer that are transferable to any discipline and any context. A range of cognitive skills, meta skills, being a design thinker and the necessity of representational skills are some elements of a designer and have been identified as being transferable across disciplines and contexts.

> "Many of the cognitive processes and like coordination processes are the same, the kind of responding to contexts, engaging with humans, engaging with technology. There are many parallels" (Hugo).

3.7 Design is teachable and learnable

A consistent element across the data is that it is possible to teach and learn to design. Students can have varying starting points in relation to their skills and abilities but there is no identification of unlearnable traits of a designer. A frequent identified element within the data is that creativity is learnable and teachable, that there are tools to support creativity, and the development of this skill. There exist differentiating levels of creativity that students can engage in, dependent on the uncertainty of the design activity, the guidance from the educator and the use of supporting tools and methods.

> "I would say that all students can design but, we have to assess them, grade them. So there is some difference and some students or humans are better at finding solution(s), but I think you can train that ability as well" (Grace).

3.8 Development of Meta Skills

Discussed in this code is the necessity and intent of current design practices to develop and foster meta skills to enable lifelong learning. A frequent skill identified is the ability to reflect, ultimately developing towards a reflective practitioner which enables the student to understand the 'why' of a design process and activity in order to apply this to other scenarios and contexts never experienced before. The uncertain nature of design demands that students can be confident in

their ability to engage in never seen before scenarios through the development and fostering of meta skills. It is frequently identified, how engagement in a design process can support the fostering of said skills for lifelong learning.

> "So my focus is really on like meta skills" (Hugo).

3.9 Developing understanding of design

The participants shared the journey that their understanding of design, designing and the implementation of design activities and tasks have developed throughout their career. This code refers to the progression of the field of design research and the impact that it has on academics within the area. Insights to this code include, participants sharing the progression of their understanding of design as they developed as academics and their journey to become experts within their field.

> "I've got a more nuanced and a more theoretical understanding of design now as an academic" (Frank).

3.10 Trapped in a process

This code refers to the participants describing how students can become trapped within a design process, particularly when the students perceive that there is a predetermined process to consider. The entrapment is described as the result of engaging students in design activities that follow a particular design process or processes, and so the moment that the student gets fixated on a design, struggles within a certain element, they find it very difficult to overcome this to design a successful solution to the task.

> "when I was younger, I suppose I felt more trapped by, well, there's the double diamond and then there's stage gate and then there's agile and then there's design thinking… I'm much more confident to look at the more like underlying mechanisms and that essentially all of these models are specific embodiments of underlying ideas" (Hugo).

4 CONCLUSION

A limitation to this paper is the small participant sample size, therefore the associated findings are not representative of the potential findings of this project. They represent a point in the process of the on-going project, to consider the initial findings and also to illustrate the iterative relationship between data collection and analysis that takes place simultaneously through this methodology. It must also be recognised that there are a multitude of codes presented in the findings that is indicative of this stage of this phase, but during the final stages of data analysis these codes and future ones will be grouped conceptually to a greater level of abstraction that will support the identification of an emerging theory.

From these initial conversations with experts within the field, confidence is seen within the remit of design being teachable and learnable, where being able to successfully design can be enhanced and fostered, and that there exist transferable skills across design contexts, ultimately fostering a whole designer, independent of time or place. Recognition towards the possibility of enhancing all students' design capabilities in technology education elicits a positive bright future for design education, ultimately making our field which sees design as a main contributor, being accessible to everyone. However, with the identified need to further enhance current pedagogical practices and make applicable the expert, tacit knowledge, top-down understanding of design theory to a more open and available level.

It is clear there is a wide-ranging complexity to design, designing and designer identified from these initial findings. Ensuring design's true nature of being free, autonomous and uncertain can provide difficulties and challenges for implementation into curricula and classroom practice. The potential of design becoming formulaic may result in students becoming trapped in a design process, not being equipped with dealing with uncertainty in design and limiting their transferability across disciplines and contexts. Williams (2000) described there to be no simple generalizable process to designing or how to teach it, which is consistent with the key findings discussed within this paper. However, what is also consistent from these findings and existing research (Hennessy & McCormick, 1994; Williams, 2000) is the problematic relationship between design and its appropriate assessment approaches. When design activity is valued to be uncertain and ill-defined but is still described to be manipulated to become defined, prescribed and closed within current assessment approaches, raises the concern as to what has progressed to reduce or resolve this issue over the past two decades. The complexity of this issue is evidently significant, but it is not unknown. One avenue of investigation uses adaptive comparative judgement which points positively towards a resolution (Buckley et al., 2020; Seery et al., 2012), however it is not yet widely adopted in practice. Will assessing design always provide core challenges to preserving design, and will this and other complexities of design always induce strain on the involvement of true design in technology education?

As this is an on-going study utilising the principles of a classical grounded theory methodology, there is a limited reference to literature and theoretical relationships to previous research conducted. Continuation of data collection and analysis using constant comparison, selective and theoretical coding, and the engagement of participants from a range of varied experiences and geographical contexts, will support the investigation of progressing designerly capability in technology education and ultimately the complexity of design.

Disclosure statement: Please note that there was no use of generative artificial intelligence (AI) tools to conduct this research or write this paper presented.

5 REFERENCE LIST

Atkinson, S. (2017). So what went wrong and why? In K. Baynes & E. Norman (Eds.), *Design epistemology and curriculum planning* (pp. 13–17). Loughborough Design Press. http://www.ldpress.co.uk/book/design-epistemology-curriculum-planning/

Baynes, K. (2006). Design Education: What's the point? *Design and Technology Education: An International Journal*,*11*(3).https://openjournals.ljmu.ac.uk/DesignTechnologyEducation/article/download/2171/1087

Buckley, J., Canty, D., & Seery, N. (2020). An exploration into the criteria used in assessing design activities with adaptive comparative judgment in technology education. *Irish Educational Studies*, *0*(0), 1–19. https://doi.org/10.1080/03323315.2020.1814838

de Vries, M. J. (2016). *Teaching about Technology: An Introduction to the Philosophy of Technology for Non-philosophers*. Springer.

Glaser, B. G., & Strauss, A. L. (1967). *The discovery of grounded theory: Strategies for qualitative research*. Aldine de Gruyter.

Glaser, B., & Strauss, A. (2017). *Discovery of Grounded Theory: Strategies for Qualitative Research*. Routledge. https://doi.org/10.4324/9780203793206

Hennessy, S., & McCormick, R. (1994). The general problem-solving prcoess in technology education: Myth or reality. In *Teaching and learning technology*. Routledge.

Kenny, M., & Fourie, R. (2015). Contrasting Classic, Straussian, and Constructivist Grounded Theory: Methodological and Philosophical Conflicts. *Qualitative Report*, *20*, 1270–1289. https://doi.org/10.46743/2160-3715/2015.2251

McDyer, M., Buckley, J., Dunbar, R., Blom, N., & Seery, N. (2022). A categorisation of the aims of design-based research in technology education. *PATT on the Edge: Teaching, Innovation and Education*. PATT on the Edge: Teaching, Innovation and Education.

McDyer, M., Buckley, J., Dunbar, R., Blom, N., & Seery, N. (2023). Problematising and unpacking the uncertainty of design within technology education. *The 40th International Pupils' Attitudes Towards Technology Conference Proceedings 2023*, https://openjournals.ljmu.ac.uk/PATT40/article/view/1531

McDyer, M., Buckley, J., Dunbar, R., Blom, N., & Seery, N. (2024). Design teaching in classroom practice: Phase 1 of a grounded theory study in technology education. *The 41st International Pupils' Attitudes Towards Technology Educational Research Conference*. PATT41.

Mitcham, C. (2001). Dasein Versus Design: The Problematics of Turning Making Into Thinking. *International Journal of Technology and Design Education*, *11*(1), 27–36. https://doi.org/10.1023/A:1011282121513

Ritz, J. M., & Martin, G. (2013). Research needs for technology education: An international perspective. *International Journal of Technology and Design Education*, *23*(3), 767–783. https://doi.org/10.1007/s10798-012-9215-7

Seery, N., Canty, D., & Phelan, P. (2012). The validity and value of peer assessment using adaptive comparative judgement in design driven practical education. *International Journal of Technology and Design Education*, *22*(2), 205–226. https://doi.org/10.1007/s10798-011-9194-0

Spendlove, D. (2015). Developing a Deeper Understanding of Design in Technology Education. In P. J. Williams, A. Jones, & C. Buntting (Eds.), *The Future of Technology Education* (pp. 169–185). Springer. https://doi.org/10.1007/978-981-287-170-1_9

Stables, P. K. (2008). Designing matters; designing minds: The importance of nurturing the designerly in young people. *Design and Technology Education: An International Journal*, *13*(3). https://ojs.lboro.ac.uk/DATE/article/view/179

Steltenpohl, C. N., Lustick, H., Meyer, M. S., Lee, L. E., Stegenga, S. M., Reyes, L. S., & Renbarger, R. L. (2024). Rethinking Transparency and Rigor from a Qualitative Open Science Perspective. *Journal of Trial & Error*, *4*(1). https://doi.org/10.36850/mr7

Walker, D., & Myrick, F. (2006). Grounded Theory: An Exploration of Process and Procedure. *Qualitative Health Research*, *16*(4), 547–559. https://doi.org/10.1177/1049732305285972

Williams, A., Cowdroy, R., & Wallis, L. (2013). Design. In *Technology Education for Teachers* (pp. 98–114). Sense Publishers. https://link.springer.com/content/pdf/10.1007/978-94-6209-161-0.pdf#page=21

Williams, P. J. (2000). Design: The Only Methodology of Technology. *Journal of Technology Education*, *11*(2). https://doi.org/10.21061/jte.v11i2.a.4

Williams, P. J. (2016). Research in Technology Education: Looking back to move Forward … again. *International Journal of Technology and Design Education*, *26*(2), 149–157. https://doi.org/10.1007/s10798-015-9316-1

Williams, P. J. (2017). *Technology Education: History of Research* (pp. 1–15). https://doi.org/10.1007/978-3-319-38889-2_21-1

Xu, M., Williams, P. J., Gu, J., & Zhang, H. (2020). Hotspots and Trends of Technology Education in the International Journal of Technology and Design Education: 2000–2018. *International Journal of Technology and Design Education*, *30*(2), 207–224. https://doi.org/10.1007/s10798-019-09508-6